

Two Complementary Signals for Systematic Index Futures Trading: Volatility-Conditioned Mean Reversion and Spline-Curvature Trend Following on Micro E-mini Nasdaq-100 Futures

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Abstract

This paper presents a two-signal systematic trading framework for Micro E-mini Nasdaq-100 (MNQ) futures and formalizes the mathematics underlying each component. The first subsystem is a volatility-conditioned mean-reversion model, motivated as a discrete-time analogue of an Ornstein–Uhlenbeck reversion process, that discretizes rolling z-scores of MNQ price displacement and CBOE Volatility Index (VIX) level into standard-deviation bins and trades joint bin combinations selected through a rolling walk-forward validation process subject to explicit statistical-significance and multiple-comparisons corrections. The second subsystem is a spline-curvature trend-following model that fits a cubic smoothing spline to a rolling window of price bars and uses the sign and magnitude of its second derivative — a direct estimate of local curvature — to identify genuine trend inflections, filtered by a long-horizon trend indicator and a cross-asset correlation check. The two subsystems are combined under an explicit priority and confluence rule, motivated by Kelly-criterion sizing logic, in which the mean-reversion signal takes precedence and agreement between the two subsystems increases position size. This paper develops the governing equations of both subsystems from first principles, formalizes the filters that gate each signal, and describes the walk-forward and Monte Carlo methodology used to guard against overfitting, including Probabilistic and Deflated Sharpe Ratio diagnostics used to flag curve-fit revisions during development. Calibrated parameter values and backtested performance figures are treated as the proprietary component of the system and are withheld; this paper documents architecture and theory rather than results.

Keywords: mean reversion, Ornstein–Uhlenbeck process, trend following, volatility regime, VIX, smoothing splines, curvature, z-score signals, walk-forward validation, Monte Carlo simulation, Kelly criterion, Probabilistic and Deflated Sharpe Ratio

1. Introduction

Systematic futures strategies are typically built around one of two broad hypotheses about short-horizon price behavior: that extreme, statistically unusual dislocations tend to revert, or that a genuine directional move tends to persist once it is underway. These hypotheses are not mutually exclusive, and this paper presents a framework built on both, applied to Micro E-mini Nasdaq-100 (MNQ) futures.

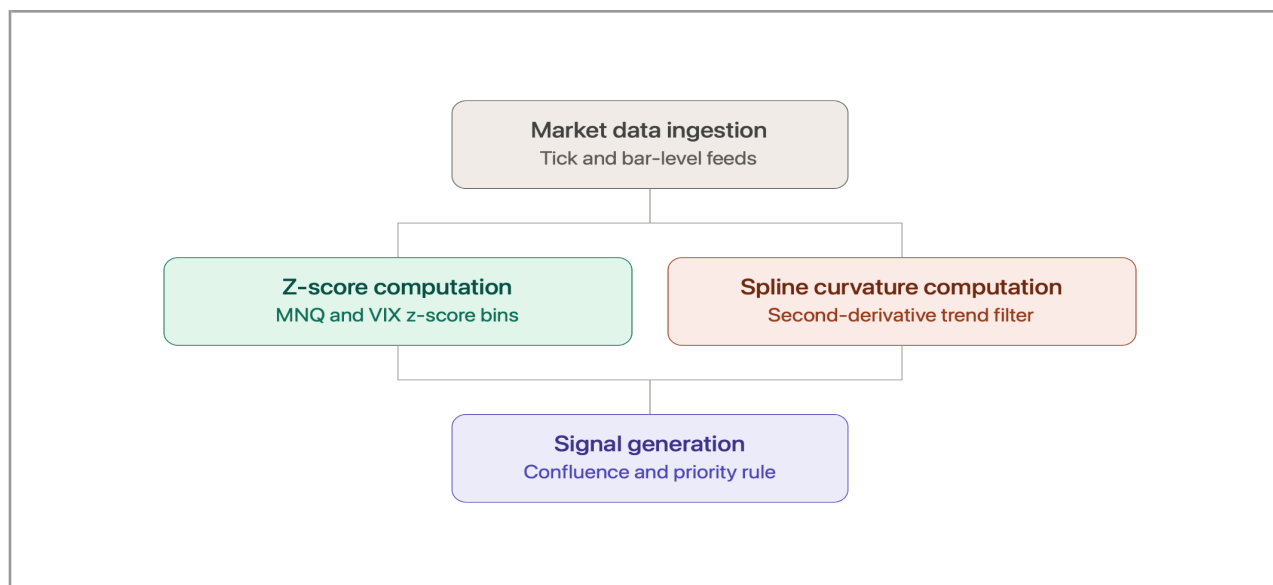
The first subsystem is a volatility-conditioned mean-reversion model. It rests on the intuition that a sharp directional move in MNQ occurring alongside an extreme reading in the CBOE Volatility Index (VIX) often reflects a transient imbalance between order flow and fair value rather than a durable repricing of risk, and that such joint dislocations carry statistically favorable reversion characteristics over a short holding horizon. The second subsystem is a

spline-curvature trend-following model. It rests on the complementary intuition that the local curvature of a smoothed price path — not merely its slope — carries information about whether a directional move is beginning, sustaining, or exhausting, and can be used as an entry and exit filter distinct from a conventional moving-average crossover.

Rather than treating these as competing strategies to be chosen between, the framework combines them under an explicit priority and confluence rule, described in Section 5. The remainder of this paper is organized as follows. Section 2 describes the instruments and data pipeline. Section 3 formalizes the volatility-conditioned mean-reversion subsystem, including its continuous-time theoretical motivation and the statistical testing applied to candidate signals. Section 4 formalizes the spline-curvature trend-following subsystem, including an equivalent penalized-regression view of spline smoothing. Section 5 describes how the two subsystems are combined into a single position, motivated by Kelly-criterion sizing logic. Section 6 covers the shared risk-management architecture. Section 7 describes the walk-forward, Monte Carlo, and overfitting-diagnostic methodology used to validate both subsystems, including a multiple-comparisons correction for the bin- and parameter-search process itself. Section 8 discusses the transition from backtest to live and paper implementation. Section 9 discusses limitations, and Section 10 concludes.

2. Instruments and Data

Both subsystems trade Micro E-mini Nasdaq-100 futures (MNQ). The mean-reversion subsystem is conditioned on the level of VIX futures (VX), sampled independently of the MNQ series. The trend-following subsystem maintains a parallel series in Micro E-mini S&P 500 futures (MES), used exclusively as a cross-asset correlation reference rather than as a tradable instrument in this framework. Price data was sourced from institutional-grade tick and bar-level futures feeds, with the mean-reversion subsystem operating on fine-grained intraday bars to support precise, edge-triggered signal detection, and the trend-following subsystem operating on a coarser, multi-hour bar aggregation more appropriate to its longer holding horizon.



3. Subsystem I: Volatility-Conditioned Mean Reversion

3.1 Theoretical Basis: Mean Reversion as an Ornstein–Uhlenbeck Process

The hypothesis that an extreme price/volatility dislocation tends to revert can be given a standard continuous-time justification via the Ornstein–Uhlenbeck (OU) process, a canonical model for a mean-reverting stochastic variable:

$$dX_t = \theta(\mu - X_t)dt + \sigma dW_t$$

Eq. 1 — Ornstein–Uhlenbeck process: X_t reverts toward its long-run mean μ at rate θ , subject to Brownian noise of intensity σ .

A defining property of the OU process is that the expected future deviation from the mean decays exponentially with the horizon h :

$$E[X_{t+h} - \mu \mid X_t] = (X_t - \mu) \cdot e^{-\theta h}$$

Eq. 2 — Expected reversion of a displaced OU process after horizon h .

Setting the right-hand side of Eq. 2 to one-half of the initial displacement and solving for h yields the process's half-life — the horizon over which half of any given displacement is expected to have decayed:

$$\tau_{1/2} = \ln(2) / \theta$$

Eq. 3 — Half-life of mean reversion.

This continuous-time framing motivates two empirical design choices developed in the sections that follow. The rolling z-score of Section 3.2 is a discrete-time, unit-free estimate of how far X_t currently sits from its estimated long-run mean relative to its own noise scale σ — the empirical analogue of the displacement term in Eq. 2. The time-stop of Section 3.7 is a pragmatic, data-driven proxy for a maximum holding horizon beyond which a position that has not reverted is statistically unlikely to do so within the process's characteristic half-life, rather than an arbitrarily chosen cutoff.

3.2 Rolling Z-Score Construction

For a price or index series X observed at time t , a rolling z-score measures how many standard deviations the current observation sits from its own trailing distribution, computed over a lookback window of w bars:

$$Z_t = (X_t - \mu_w) / \sigma_w$$

Eq. 4 — Rolling z-score, where μ_w and σ_w are the trailing sample mean and standard deviation of X over the most recent w bars.

This construction is applied independently to two series: the MNQ price series, whose z-score captures short-horizon directional displacement, and the VIX level series, whose z-score captures how extended the current level of implied volatility is relative to its own recent regime. Both z-scores are recomputed continuously as new bars arrive.

3.3 Standard-Deviation Bin Discretization

Rather than acting on a single continuous z-score threshold, each z-score is discretized into a bin using a fixed set of standard-deviation edges:

$$\text{bin}(Z_t) = k \text{ such that } b_{k-1} \leq Z_t < b_k$$

Eq. 5 — Bin assignment, with edge set $b = \{-\infty, -2, -1, 0, 1, 2, \infty\}$ defining six bins in units of standard deviation.

Discretizing into bins rather than thresholding continuously has two advantages. First, every joint MNQ-bin $\times$ VIX-bin combination becomes independently inspectable for historical win rate, average return, and sample size, rather than assuming the relationship between z-score and forward return is linear or monotonic across the full distribution. Second, it allows the walk-forward process of Section 3.6 to select only the specific combinations with a demonstrated, out-of-sample edge, rather than a single globally fitted threshold.

3.4 Joint Signal Construction and Edge-Triggering

A trade signal fires when the MNQ bin and the VIX bin are jointly members of a qualifying combination set, identified through the walk-forward process described below:

$$\text{Enter}_t = 1\{\text{bin}_{\text{MNQ}}(t) \in S\} \cdot 1\{\text{bin}_{\text{VIX}}(t) \in V\}$$

Eq. 6 — Joint entry indicator, where S and V are the qualifying MNQ-bin and VIX-bin sets identified by walk-forward calibration (Section 3.6). A mirrored condition, with its own qualifying sets, governs short entries.

Entries are edge-triggered: the signal fires on the first bar on which the joint bin condition becomes true, rather than on every bar for which it continues to hold. This is a deliberate design choice intended to capture the reversion trade as close as possible to the point of maximum dislocation, rather than at some arbitrary later point after the condition has already persisted for some time.

3.5 Statistical Significance of Bin Combinations

Before a bin combination is admitted to the walk-forward selection procedure of Section 3.6, its historical edge must be statistically distinguishable from noise. For a bin combination k observed n_k times in a lookback window, with mean forward return $\bar{r}_k$ and sample standard deviation s_k , this is tested via a standard one-sample t-statistic against a null hypothesis of zero mean return:

$$t_k = \bar{r}_k / (s_k / \text{sqrt}(n_k))$$

Eq. 7 — t-statistic for the significance of bin combination k 's average forward return, under $H_0: \mu_k = 0$.

Because the bin-matrix search evaluates many combinations simultaneously, a single combination clearing an uncorrected significance threshold is weak evidence on its own — with a large enough grid, some combinations will appear significant by chance alone. A Bonferroni-style correction adjusts the per-comparison significance level for the number of combinations M being tested simultaneously:

$$\alpha_{\text{adj}} = \alpha / M$$

Eq. 8 — Bonferroni-adjusted significance threshold for M simultaneously tested bin combinations.

This multiple-comparisons correction is a necessary companion to bin-matrix search, and is applied again at the strategy-configuration level in Section 7.3 via the Deflated Sharpe Ratio — the same underlying concern (many trials inflate the chance of an apparently significant result) recurs at both the individual-signal level and the overall-configuration level, and is treated explicitly at each.

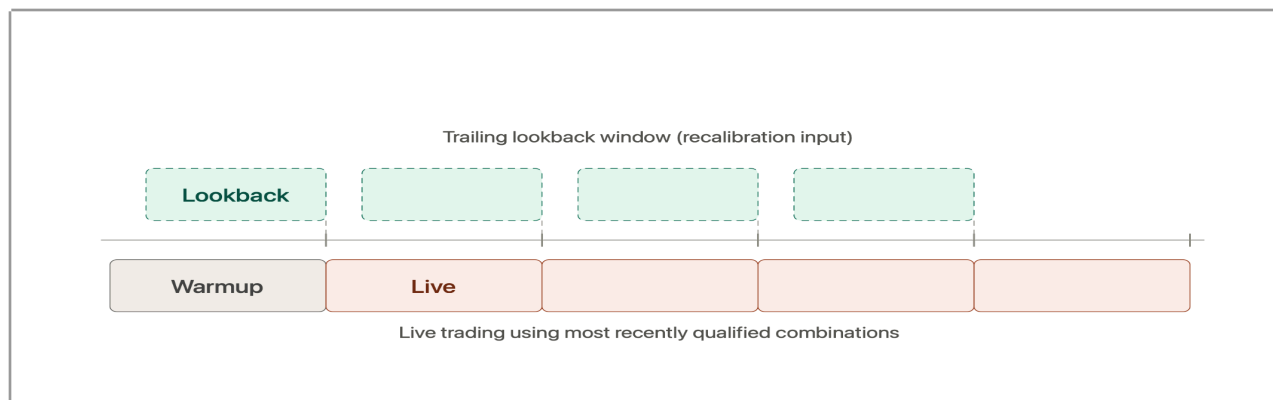
3.6 Walk-Forward Calibration

A central methodological concern in any bin-search strategy is overfitting: with enough degrees of freedom, it is straightforward to find a bin combination that performed well over a fixed historical sample purely by chance. To guard against this, the qualifying sets S and V in Eq. 6 are not chosen via a single in-sample optimization over the full dataset, but through a rolling walk-forward procedure:

$$(S, V)_m = \operatorname{argmax} \operatorname{Perf}((S, V); W_m)$$

Eq. 9 — Monthly recalibration, where $W_m = [t_m - L, t_m]$ is a trailing lookback window ending at the start of month m , and $\operatorname{Perf}(\cdot)$ is a joint win-rate / significance / sample-size objective incorporating the t -statistic of Eq. 7.

An initial warmup period is used to build stable z-score distributions before any calibration begins. Thereafter, the qualifying bin sets are recalibrated on a rolling basis using the trailing lookback window W_m , and a given combination is only carried forward into the next out-of-sample trading window if it continues to meet minimum win-rate, significance, and sample-size thresholds. This ensures that every simulated trade is generated using only information available prior to that trade, and that the combinations ultimately used in live trading are those that repeatedly re-qualified across many independent out-of-sample windows rather than a single lucky in-sample fit.



3.7 Adaptive Exit and Time-Stop

Each qualifying bin combination carries its own adaptive exit bin — the z-score bin at which the position is closed on favorable reversion — rather than a single fixed profit target applied uniformly, since the expected magnitude of reversion differs by how extended the entry condition was. For losing trades, a dedicated time-stop is enforced:

$$Exit_{loss}(t) = 1\{(t - t_0) > T_{max}\} \cdot 1\{PnL_t < 0\}$$

Eq. 10 — Time-stop rule: a losing position open longer than T_{max} bars is closed regardless of the adaptive exit-bin condition.

This rule was motivated by holding-time analysis showing that losing trades still open well past a typical winning trade's holding time were disproportionately weekend-adjacent holds unlikely to revert within a reasonable horizon — consistent with the half-life reasoning of Eq. 3, under which a position that has failed to revert well beyond $\tau_{1/2}$ is behaving inconsistently with the OU hypothesis that motivated the trade. Enforcing a time-stop on losers materially reduces capital tied up in stagnant positions with only a small cost to eventual winners.

4. Subsystem II: Spline-Curvature Trend Following

4.1 Motivation: Why Curvature

A price path that is genuinely turning — beginning a new trend leg — looks locally like the bottom or top of a bowl: it curves in a consistent direction over several bars, not merely moving. A price path that is noisy chop, by contrast, wiggles back and forth with plenty of first-derivative (slope) variation but little persistent second-derivative (curvature) structure. This motivates curvature, rather than slope alone, as an entry filter: slope indicates that price moved, curvature indicates the shape of how it moved.

Raw tick- or minute-level price data is too jagged to differentiate directly — a numerical second derivative of raw closes is dominated by microstructure noise and is not usable as a signal on its own. The role of a smoothing spline is to first produce a smooth, twice-differentiable estimate of the underlying price path, from which a stable second derivative can then be computed.

4.2 Cubic Smoothing Splines

A cubic smoothing spline $s(x)$ fit to a set of price observations (x_i, y_i) is the curve that best balances two competing goals: staying close to the observed prices, and remaining smooth. This tradeoff is controlled by a single smoothing budget that upper-bounds the total deviation of the fitted curve from the raw data:

$$\sum_{i=1}^n (y_i - s(x_i))^2 \leq S$$

Eq. 11 — Smoothing constraint bounding total fit error by the smoothing budget S .

An equivalent and often more illuminating formulation expresses the same tradeoff as a single penalized objective, in which the fitted curve directly minimizes a weighted combination of fit error and curvature:

$$s = \operatorname{argmin}_g [\sum_{i=1}^n (y_i - g(x_i))^2 + \lambda \cdot R(g)]$$

Eq. 12 — Penalized (Reinsch) form of the smoothing spline objective, equivalent to the constrained form of Eq. 11 for a corresponding λ , where $R(g)$ is the integrated squared curvature penalty $\int g''(x)^2 dx$.

This penalized form makes explicit something only implicit in Eq. 11: curvature is not merely computed after the fact from the fitted spline — it is the literal quantity being suppressed by the

fitting procedure itself. The larger λ , the more the objective penalizes curvature, and the smoother and less curvature-sensitive the resulting $s''(x)$ becomes; conversely a small λ permits the fit to bend more freely, producing a more reactive but noisier second derivative. Tuning the smoothing parameter is therefore, quite literally, tuning how much curvature the entry signal of Section 4.3 is permitted to detect. The spline is piecewise cubic, which guarantees it is twice continuously differentiable everywhere — a necessary property, since the trading signal is built directly from its second derivative.

4.3 Curvature as the Second Derivative

Before considering the analytic spline formulation, it is useful to note that the second derivative of a discretely sampled path can be estimated directly from a central finite difference:

$$s''(x) \approx (s(x+h) - 2s(x) + s(x-h)) / h^2$$

Eq. 13 — Central finite-difference approximation to the second derivative.

This view is useful for intuition — it shows curvature as literally measuring how far the middle point sits from the average of its two neighbors, i.e. how much the path bends locally — even though the deployed implementation instead evaluates the analytic derivative of the fitted piecewise-cubic spline directly at each bar, avoiding the noise amplification that differencing raw or lightly-smoothed price data would introduce. For a smooth curve $s(x)$, the formal geometric measure of signed curvature at a point accounts for both this second derivative and the local slope:

$$\kappa(x) = s''(x) / (1 + s'(x)^2)^{3/2}$$

Eq. 14 — Signed curvature of a smooth price path.

In the small-slope regime typical of short-horizon futures price action, the denominator in Eq. 14 stays close to one, so the raw second derivative $s''(x)$ alone already carries the essential shape information: its sign indicates whether the price path is curving upward (concave up, $s'' > 0$) or downward (concave down, $s'' < 0$), and its magnitude indicates how sharply. This is the practical basis for using $s''(x)$, evaluated at the most recent bar, as the core trading signal. A long entry requires this curvature estimate to exceed a minimum positive threshold, rather than simply being greater than zero, so that only meaningfully pronounced inflections are treated as tradable:

$$s''(x_t) > \kappa_{min}$$

Eq. 15 — Minimum-curvature entry gate.

4.4 Trend Filter

The curvature signal identifies a local inflection but says nothing about the broader directional backdrop, so a long-horizon exponential moving average (EMA) trend filter is layered on top, requiring price to sit above its long-horizon EMA before a curvature-based entry is permitted:

$$EMA_t = \alpha \cdot P_t + (1-\alpha) \cdot EMA_{t-1}, \quad \alpha = 2/(N+1)$$

Eq. 16 — EMA recursion over an N-bar horizon.

Unrolling this recursion shows the EMA to be equivalent to an infinite weighted sum of past prices, with weights decaying geometrically into the past:

$$EMA_t = \alpha \cdot \sum_{k=0}^{\infty} (1-\alpha)^k \cdot P_{t-k}$$

Eq. 17 — Closed-form expansion of the EMA recursion: an exponentially-weighted moving average with weights decaying geometrically at rate $(1-\alpha)$.

This form makes explicit why the EMA behaves as a smoothed, lagged trend estimate: it weights recent prices most heavily while never fully discarding older observations, in contrast to a simple moving average's hard cutoff at the edge of its window. This restricts curvature-based entries to inflections occurring in the context of an already-established or re-emerging uptrend, rather than treating a curvature inflection inside a broader downtrend as a long signal.

4.5 Cross-Asset Correlation Filter

A rolling correlation check against MES is also enforced, requiring recent MNQ and MES returns to be sufficiently correlated before an entry is permitted. The correlation itself is the standard Pearson correlation coefficient between the two return series over the rolling window:

$$\rho(r_{MNQ}, r_{MES}) = \text{Cov}(r_{MNQ}, r_{MES}) / (\sigma_{MNQ} \cdot \sigma_{MES})$$

Eq. 18 — Pearson correlation coefficient: the ratio of the covariance between the two return series to the product of their standard deviations.

The entry filter requires this quantity to clear a minimum threshold:

$$\rho(r_{MNQ}, r_{MES}) \geq \rho_{min}$$

Eq. 19 — Cross-asset correlation gate.

The intent is to filter out entries driven by idiosyncratic, Nasdaq-100-specific moves that are not corroborated by the broader S&P 500 futures market, on the premise that broad-based moves are more likely to reflect a genuine macro or index-level trend shift than single-sector noise.

4.6 Curvature-Reversal Exit

The same quantity that gates entry also governs exit. A long position opened on positive curvature is held while the smoothed path continues to curve upward, and closed when the sign of $s''(x)$ flips negative — the geometric statement that the concave-up phase of the move has ended:

$$\text{sgn}(s''(x_t)) \neq \text{sgn}(s''(x_{\text{entry}}))$$

Eq. 20 — Curvature-reversal exit condition, subject to a minimum holding period before it is permitted to act.

This design is internally consistent — the trade's entry thesis and its termination condition derive from the same measurement — and adaptive: in a strong, extended trend the curvature can remain favorably signed for a long time, letting winners run, while a move that quickly exhausts triggers a correspondingly quick exit. A minimum holding period is enforced before this exit is permitted to act, to prevent immediate churn when curvature hovers near zero at the moment of entry.

5. Signal Combination: Priority and Confluence

5.1 Theoretical Motivation for Confluence-Based Sizing

The decision to scale position size on confluence rather than treat all qualifying signals identically has a standard theoretical grounding in the Kelly criterion, which characterizes the growth-optimal fraction of capital to risk on a favorable bet as a function of the bet's edge:

$$f^* = p - (1-p)/b$$

Eq. 21 — Kelly-optimal fraction f^ for a bet with win probability p and payout ratio b (win/loss magnitude ratio). Growth-optimal sizing increases monotonically with the estimated edge p .*

Confluence between two independently derived signals is naturally read as increasing the effective probability that a position is correctly directed, relative to either signal in isolation — in the same qualitative sense that a higher p increases the Kelly-optimal fraction in Eq. 21. The sizing rule that follows should be read in this spirit, as a discrete, conservative approximation to Kelly-consistent sizing rather than a fully optimized fractional-Kelly implementation; a formal calibration of position size against estimated confluence edge is discussed as future work in Section 9.

5.2 Priority and Sizing Rule

Let $L_Z(t), S_Z(t) \in \{0,1\}$ indicate an active long or short mean-reversion signal from Subsystem I at time t , and let $L_S(t) \in \{0,1\}$ indicate an active long trend-following signal from Subsystem II. The combined position, in units of a base contract size n_{base} is governed by a priority rule in which the mean-reversion subsystem dominates:

$$n(t) = n_{base} \cdot (1 + L_Z(t) \cdot L_S(t)), \text{ if } S_Z(t) = 0$$

$$n(t) = -n_{base} \text{ if } S_Z(t) = 1$$

Eqs. 22–23 — Confluence position-sizing rule. Agreement between both subsystems on a long position scales size above the base allocation; an active mean-reversion short signal overrides and flips the position regardless of the trend-following subsystem's state.

Two properties of Eqs. 22–23 are worth making explicit. First, the mean-reversion subsystem's short signal is treated as higher-priority information than any long-side trend-following state — the rationale being that an extreme joint price/volatility dislocation is a stronger, more specific statistical statement than a curvature inflection. Second, confluence — both subsystems independently agreeing on a long position — is treated as corroborating evidence and rewarded with increased size, rather than the two signals being blended into some intermediate continuous weighting. This keeps the combination rule fully interpretable: at any point in time, the position held can be attributed to a specific, human-readable combination of subsystem states.

6. Risk Management Architecture

Both subsystems share a common risk-control philosophy, layered around each subsystem's own primary exit logic (the adaptive exit bin and time-stop of Section 3.7 for Subsystem I; the curvature-reversal exit of Section 4.6 for Subsystem II):

- A hard stop-loss is submitted as a real, broker-resting stop order at entry rather than evaluated only inside the algorithm's own loop, so that protection persists even if the algorithm's evaluation is delayed, disconnected, or mid-contract-roll.
- Partial profit-taking locks in a portion of unrealized gains on a sufficiently favorable excursion, letting the remainder ride to the subsystem's primary exit.
- A minimum holding period is enforced before either primary exit condition is permitted to act, preventing immediate churn from noise near the moment of entry.
- A session/weekend flat rule closes all open positions ahead of periods of materially reduced liquidity or elevated gap risk, so that no position is carried into a non-trading window without an explicit, separately validated multi-day holding thesis.

The trend-following subsystem is deployed long-only: development-stage attribution analysis showed that short-side entries under the curvature signal were responsible for a disproportionate share of realized drawdown relative to their contribution to performance, and the short side was subsequently removed from the live implementation.

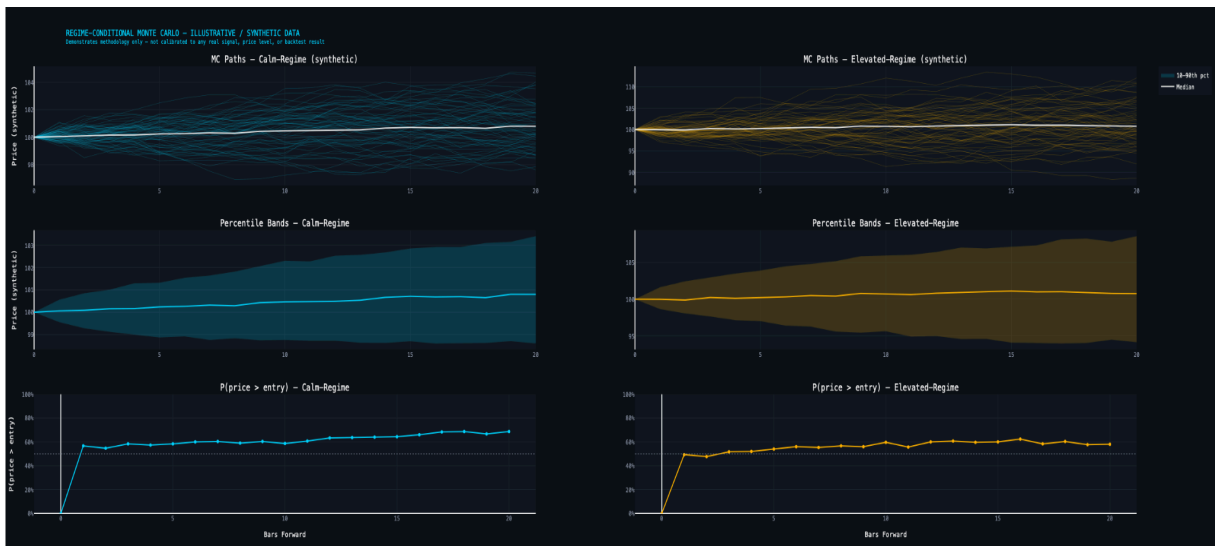
7. Validation Methodology and Overfitting Safeguards

7.1 Walk-Forward and Regime-Conditional Monte Carlo

Beyond the walk-forward calibration of Section 3.6, the robustness of the combined system's return profile is further assessed via a regime-conditional Monte Carlo simulation. Rather than resampling trade outcomes from a single pooled distribution, simulated returns are drawn from empirical distributions conditioned jointly on the specific signal state (bin combination or curvature regime) and the prevailing volatility regime at the time of each simulated trade:

$$\tilde{r}_k \sim \hat{F}(\cdot \mid \text{bin}_k, \text{regime}_k)$$

Eq. 24 — Regime-conditional resampling: each simulated trade return is drawn from the empirical distribution conditioned on its originating bin/regime, preserving the conditional structure of the strategy's edge rather than assuming i.i.d. outcomes across all market conditions.



7.2 Probabilistic Sharpe Ratio Diagnostic

Iterating a strategy across many backtested revisions on a finite historical sample carries an inherent risk of curve-fitting: it is possible to find a rule variant that performed well purely because it happened to fit the idiosyncrasies of that specific sample. A useful diagnostic for this, used during development to flag such revisions, is the Probabilistic Sharpe Ratio (PSR), which estimates the probability that a strategy's true Sharpe ratio exceeds a benchmark τ , correcting for the non-normality of a finite return sample:

$$PSR(\tau) = \Phi \left((SR - \tau) \cdot \sqrt{n-1} / \sqrt{1 - \gamma_3 \cdot SR + ((\gamma_4 - 1)/4) \cdot SR^2} \right)$$

Eq. 25 — Probabilistic Sharpe Ratio (Bailey and López de Prado, 2012). Φ is the standard normal CDF, SR the backtested Sharpe ratio, n the number of observations, and γ_3, γ_4 the sample skewness and kurtosis of returns.

A revision with an attractive headline backtest but a low PSR — particularly when combined with a low trade count and realized profit concentrated in a narrow sub-period rather than distributed broadly across the sample — is treated as a likely curve-fit artifact rather than a robust edge, and is not carried forward into live deployment. This diagnostic played a direct role in rejecting at least one full strategy rewrite during development of the trend-following subsystem, despite its headline backtest appearing attractive on its face.

7.3 Accounting for Multiple Trials: The Deflated Sharpe Ratio

The PSR of Eq. 25 evaluates a single backtested Sharpe ratio against a fixed benchmark τ . But both subsystems' development involved searching over many candidate configurations — bin combinations in the mean-reversion subsystem's walk-forward grid (Section 3.5), and smoothing and filter parameters across the trend-following subsystem's revision history — which itself inflates the expected best-observed Sharpe ratio under pure chance, mirroring the multiple-comparisons concern already raised for individual bin significance in Eq. 8. The Deflated Sharpe Ratio (DSR) corrects for this by first estimating the expected maximum Sharpe ratio one would observe after N independent trials even under a null of zero true skill, using an extreme value theory approximation:

$$E[\max_N SR] \approx \sigma_{SR} \cdot [(1-\gamma) \cdot \Phi^{-1}(1-1/N) + \gamma \cdot \Phi^{-1}(1-1/(N \cdot e))]$$

Eq. 26 — Expected maximum Sharpe ratio across N independent trials under the null of no true skill (Bailey and López de Prado, 2014), where σ_{SR} is the standard deviation of the trial Sharpe ratios and $\gamma \approx 0.5772$ is the Euler-Mascheroni constant.

The Deflated Sharpe Ratio is then the PSR of Eq. 25 evaluated using this trial-adjusted quantity as the benchmark, rather than a fixed $\tau = 0$:

$$DSR = PSR(E[\max_N SR])$$

Eq. 27 — Deflated Sharpe Ratio: PSR evaluated against the expected-under-chance maximum Sharpe ratio for N trials, penalizing configurations selected from a large search space.

In practice, this means a configuration surviving many rounds of bin-combination or parameter search is held to a correspondingly higher bar than one evaluated in isolation — directly formalizing the informal caution, discussed throughout this paper, against treating an attractive backtest from a heavily iterated development process at face value.

7.4 Execution-Layer Auditing

Both subsystems were also subjected to execution-layer auditing distinct from evaluating signal quality: re-running backtests from randomized start dates and inspecting individual trades whose realized losses exceeded what the risk architecture of Section 6 should have permitted. This practice surfaced and led to correction of defects invisible to standard full-sample backtesting, including asynchronous order-fill handling, unsafe position liquidation across futures contract-roll boundaries, and gating logic that could, under certain state transitions, fail to arm protective orders. The general lesson is that signal logic tends to fail loudly — bad trades show up directly in the equity curve — while execution logic tends to fail quietly, in ways that aggregate backtest statistics average over; targeted auditing around contract rolls and expiry dates is comparatively inexpensive and caught real defects here.

8. Live and Paper Implementation

Both subsystems were developed and backtested on QuantConnect before transitioning to a paper-trading implementation on a retail futures brokerage platform, seeding rolling signal state directly from the same research data pipeline and loading a calibration artifact generated by the walk-forward process, so that the bin thresholds and filter parameters validated in research are exactly those used to generate live signals. This design choice avoids a common failure mode in systematic trading — a quiet divergence between the research/backtest signal logic and the production signal logic — by sharing the calibration artifact directly between the two environments rather than re-implementing it. A companion real-time visualization tool mirrors each subsystem’s internal state (z-scores, bin classification, fitted spline, and curvature series) against the live market data stream, supporting ongoing comparison of live signal behavior against backtest-implied expectations. Live capital deployment followed an extended paper-trading validation period and began deliberately small, with position size scaling gated on realized account growth rather than backtest projections.

9. Discussion and Limitations

- The mean-reversion subsystem’s edge is explicitly regime-conditional (extreme joint VIX/MNQ dislocations, Eq. 6); prolonged periods without qualifying bin crossings produce extended flat periods with no trading activity in that subsystem.
- The trend-following subsystem uses the raw second derivative $s''(x)$ (Eq. 15) as a practical proxy for the full curvature expression of Eq. 14; this small-slope approximation is standard but would need revisiting for instruments or timeframes with materially steeper characteristic price slopes.
- The trend and correlation filter thresholds of Section 4 (Eqs. 16–19) are currently fixed rather than recalibrated on a rolling out-of-sample basis, unlike the walk-forward-calibrated bin sets of Subsystem I; extending walk-forward calibration to Subsystem II’s filters is a natural direction for future work.
- The confluence rule of Eqs. 22–23 was motivated qualitatively by Kelly-criterion reasoning (Eq. 21) rather than derived as a formal solution to a fractional-Kelly sizing problem; a systematic study of alternative combination and sizing rules, potentially calibrating f^* directly against confluence-conditional win rates, is a further extension.

- The Bonferroni correction of Eq. 8 and the Deflated Sharpe Ratio of Eqs. 26–27 both assume approximate independence across trials, which may understate the true multiple-comparisons penalty when candidate bin combinations or strategy revisions are highly correlated with one another, as is often the case in practice.
- Sample size per bin combination and per curvature-regime bucket, while validated across multiple walk-forward windows, remains limited relative to institutional-scale statistical standards; continued monitoring of live performance against backtest and Monte Carlo expectations is necessary before further scaling.

10. Conclusion and Future Work

This paper formalized a two-signal systematic trading framework for MNQ futures, developing from first principles the mathematics of a volatility-conditioned mean-reversion subsystem — grounded in Ornstein–Uhlenbeck reversion theory and validated with explicit statistical-significance and multiple-comparisons corrections — and a spline-curvature trend-following subsystem built on the second derivative of a smoothed price path, shown equivalently as a penalized-regression curvature suppression problem, as a unified entry and exit signal. The two subsystems were combined under an explicit, interpretable priority and confluence rule motivated by Kelly-criterion sizing logic, and validated using a combination of walk-forward out-of-sample testing, regime-conditional Monte Carlo simulation, and Probabilistic and Deflated Sharpe Ratio diagnostics intended to guard against curve-fitting during iterative development.

Future work includes walk-forward calibration of the trend-following subsystem’s currently fixed filter thresholds, a formal fractional-Kelly search over alternative confluence and position-sizing rules, extension of the regime-conditional Monte Carlo framework to jointly model both subsystems’ correlated exposure, relaxing the independence assumption underlying the multiple-comparisons corrections of Section 7, and continued live-performance tracking to formally compare paper- and live-trading results against backtest- and simulation-implied expectations as a longer track record accumulates.

Research code and strategy implementations referenced in this paper are maintained at github.com/apollostivala-del/Quant-Research.